When to Use

To compare means when the research design includes both between-groups (independent samples) and within-groups (repeated measures or matched samples) factors. This design allows examination of two main effects and one interaction effect.

Research Hypothesis: Critics scores and audience scores differ between Superman portrayals with small and big age differences between the lead actors, and the pattern of difference between score types depends on the age difference category.

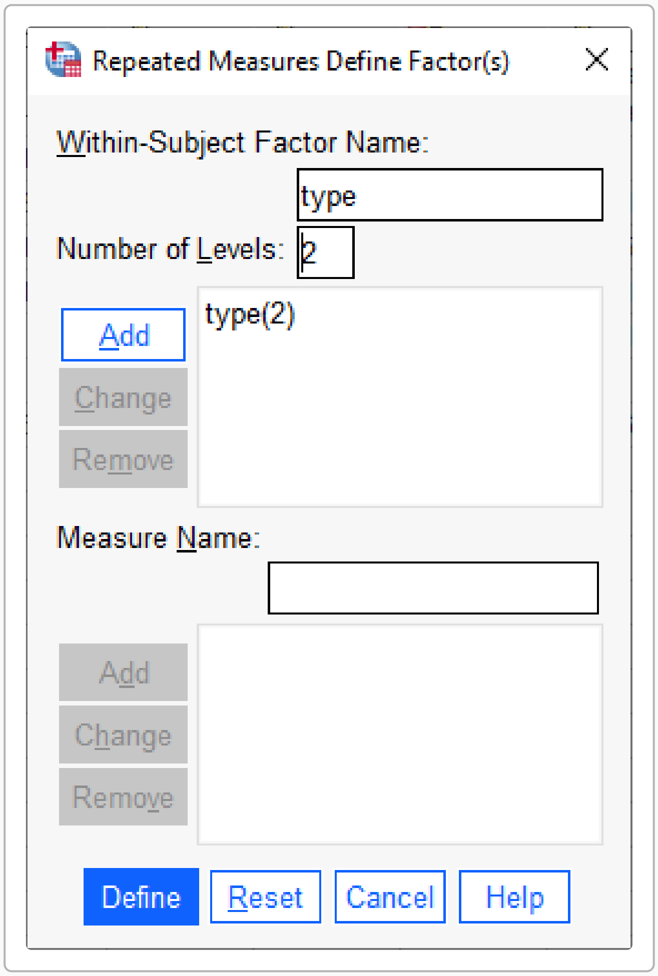
Null Hypothesis: There is no main effect of age difference category on Rotten Tomatoes scores, no main effect of score type on ratings, and no interaction between age difference and score type.

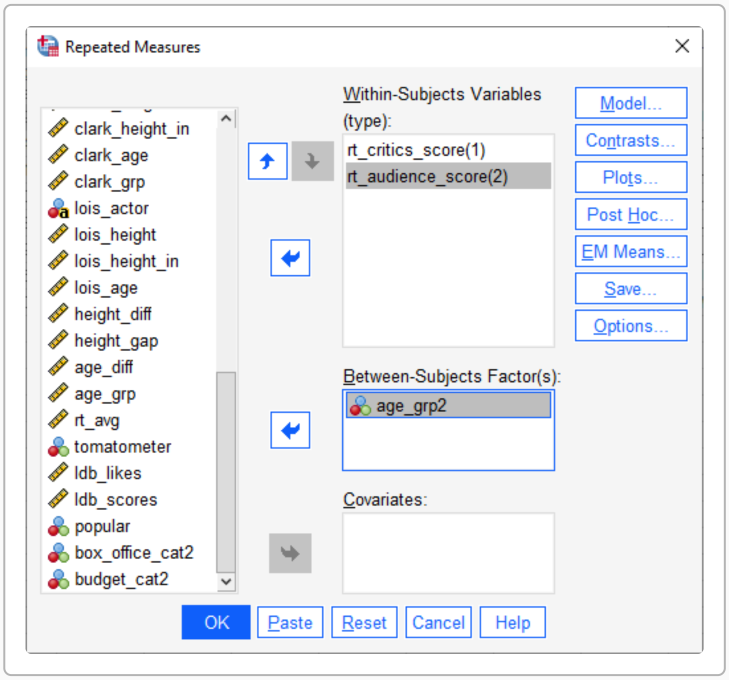
This is a **2×2 mixed factorial design** where age difference (Small vs Big) is a between-groups factor and score type (Critics Score vs Audience Score) is a within-groups factor. The **BG factor** (age_grp2) groups portrayals by age difference between Superman and Lois Lane actors — Small (under 5 years) vs Big (over 5 years). The **WG variables** (rt_critics_score, rt_audience_score) are the DV scores for each rating type.

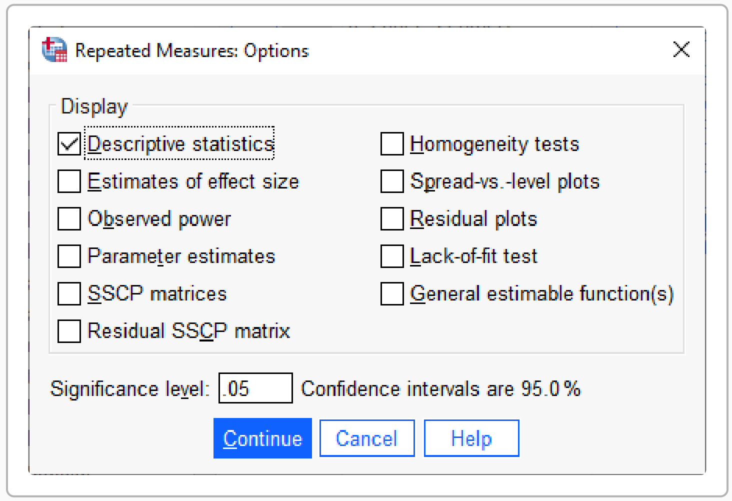
	Critics Score	Audience Score
Small		
Big		

Analyze → General Linear Model → Repeated Measures

- In the **Repeated Measures Definition Window** enter your name for the WG IV in the “Within-subject Factor Name” box (ScoreType)
- Enter the number of conditions of the WG IV in the “Number of levels” window (2)
- Click the “Add” button, then “Define”
- In the **Repeated Measures** window highlight the variables that are the DV score for each WG condition and click the arrow
- In the “Fixed Factor(s)” box, add the between-groups IV (age_grp2)
- Click “Options” — check the “Descriptives” box







SPSS Syntax

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```
GLM rt_critics_score rt_audience_score BY age_grp2   ①
  /WSFACTOR=ScoreType 2                              ②
  /PRINT=DESCRIPTIVE                                 ③
  /CRITERIA=ALPHA(.05).                              ④
```

1. List WG variables "BY" the BG IV
2. Give a name to the WG IV & tell # conditions
3. Get descriptive stats
4. Sets alpha level for significance testing

SPSS Output

Descriptive Statistics

	Mean	Std. Deviation	N
Critics Score (Small)	81.40	6.465	5
Audience Score (Small)	78.80	12.538	5

Descriptive Statistics

	Mean	Std. Deviation	N
Critics Score (Big)	76.00	16.643	3
Audience Score (Big)	83.00	7.550	3

Tests of Between-Subjects Effects

Measure: MEASURE_1 — Transformed Variable: Average

Source	Type III Sum of Squares	df	Mean Square	F	Sig.
Intercept	102080.250	1	102080.250		
age_grp2	1.350	1	1.350	0.007	.938
Error	1243.400	6	207.233		

The mixed factorial ANOVA produces three sets of results: **Descriptive Statistics** (means and SDs for each cell), **Tests of Within-Subjects Effects** (main effect of WG factor and the BG × WG interaction), and **Tests of Between-Subjects Effects** (main effect of the BG factor). The "Sphericity Assumed" row is used for within-subjects tests. The interpretation focuses on three effects: (1) **Main Effect of Age Difference (BG):** do scores differ between small vs. big age difference portrayals, averaging across score types? (2) **Main Effect of Score Type (WG):** do critics scores differ from audience scores, averaging across age groups? (3) **Interaction:** does the pattern of critics vs. audience scores depend on the age difference category? If the interaction is significant, focus on describing the cell means.

Reporting the Results

Sample APA Write-Up

A 2 x 2 mixed factorial ANOVA was conducted to examine the effects of age difference category (Small vs. Big; between-groups) and score type (critics score vs. audience score; within-groups) on Rotten Tomatoes ratings of Superman portrayals. The cell means were: Small-rt_critics_score (*M* = 81.40, *SD* = 6.47), Small-rt_audience_score (*M* = 78.80, *SD* = 12.54), Big-rt_critics_score (*M* = 76.00, *SD* = 16.64), Big-rt_audience_score (*M* = 83.00, *SD* = 7.55). The within-subjects main effect of score type was not significant, $F(1,6) = 0.49$, $p = 0.509$, $MSe = 36.77$. The between-subjects main effect of age difference was not significant, $F(1,6) = 0.01$, $p = 0.938$, $MSe = 207.23$. The interaction between age difference and score type was not significant, $F(1,6) = 2.35$, $p = 0.176$.

Key Reporting Elements

When reporting a 2x2 mixed factorial ANOVA, include:

- The research hypothesis (if there is one)
- The between-groups factor and its levels
- The within-groups factor and its levels
- Descriptive statistics for each cell (M, SD)
- Results of both main effects (*F*, *df*, *p*, *MSe*)
- Results of the interaction (*F*, *df*, *p*, *MSe*)
- Post hoc comparisons if main effects or interaction are significant
- Whether results support the research hypothesis